

Group Assignment 2: A Bayesian Subgroup Comparison of the French Short Boredom Proneness Scale

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Information about the assignment

In this group assignment, you will **reanalyze an existing dataset** using the Bayesian graphical modeling methodology for group comparisons that you just learned. You are provided with a short study description and a **specified Methods section**. Your task is to:

- Complete the definition of the inclusion Bayes factor in the Methods section
- Implement the analysis exactly as implied by the Methods section
- Write a very brief **Results section** where the results are summarized.

You will work with the **French subsample** ($N = 490$) from the original study, using **18 ordinal items** from the **Short Boredom Proneness Scale (SBPS)** and the **Curiosity and Exploration Inventory–II (CEI-II)**.

Introduction

Boredom proneness has been conceptualized as a trait-like tendency to experience boredom frequently and intensely, and has been linked to a broad range of psychological outcomes. The Short Boredom Proneness Scale (SBPS) was developed as a concise unidimensional measure of boredom proneness and has since been translated into multiple languages. In the original validation of the French version of the SBPS, Martarelli et al. (2022) examined the relationships between boredom proneness (SBPS), and measures on two subscales of the Curiosity and Exploration Inventory (CEI-II), using a regularized frequentist network approach. Their analysis suggested positive associations between boredom and curiosity items and negative associations between boredom and exploration items. However, the authors did not investigate whether the estimated network structure differs between female and male participants.

The aim of the present study is to reanalyze the French subsample of the original dataset using Bayesian graphical modeling with the specific focus on testing whether the network structure differs between female and male participants.

Method

Data

The data consists of responses from $N = 490$ French-speaking participants ($M_{age} = 33.8$; 73% females and 27% males). The network includes 18 items, all measured on a 7-point Likert scale:

- 8 items from the Short Boredom Proneness Scale (SBPS)
- 10 items from the Curiosity and Exploration Inventory–II (CEI-II)

Missing data were handled by listwise deletion.

Model Specification

We used the multi-group ordinal Markov random field (OMRF; Marsman, van den Bergh, et al., 2025; Marsman, Waldorp, et al., 2025) to model group differences in network structure. The model represents each item as a node in an undirected network, with edges encoding conditional dependence between item pairs while controlling for all variables. The multi-group extension allows the OMRF parameters to vary across groups (Marsman, Waldorp, et al., 2025).

Bayesian model averaging was used to integrate over uncertainty about which group differences were present or absent. Inference was therefore based on the posterior distribution, which

accounts for uncertainty in the OMRF parameters, the presence or absence of group differences, and, when present, the magnitude of those differences.

Prior Distributions

Structure Priors

The inclusion indicators for edge differences were assigned Beta–Bernoulli(1, 1) priors, which induce a uniform prior over the number of included differences and reflect prior ignorance about their prevalence. To assess sensitivity to this assumption, we repeated the analysis using Beta–Bernoulli(1, 10) and Beta–Bernoulli(10, 1) priors, which a priori favor few (1/11) and many (10/11) differences, respectively.

Parameter Priors

The difference parameters were assigned independent Cauchy priors with a scale of 1.0.

Inference

We quantified the evidence for the presence or absence of individual edge differences using the inclusion Bayes factor. The inclusion Bayes factor _____. These Bayes factors were used to classify edge differences as showing evidence for _____, evidence for _____, or _____ evidence.

_____ plots were used to visualize which edge differences fall into each of these categories. In this study, an inclusion Bayes factor of _____ was used as the threshold for strong evidence for inclusion, and a Bayes factor of _____ was used as the threshold for strong evidence for exclusion; values falling between these thresholds were classified as inconclusive.

Software

We used the `bgms` and `easybgm` R packages (Huth et al., 2024) to estimate the model and plot the results.

We ran the model using an adaptive Metropolis–Hastings algorithm, with all other sampling settings left at their default values. Convergence was assessed by inspecting the Gelman–Rubin statistic $\hat{R}$ for the sampled parameters. Overall, convergence was _____, with _____ parameters being below the recommended threshold of 1.05.

Results

Evidence for Edge Differences

References

- Huth, K., Keetelaar, S., Sekulovski, N., van den Bergh, D., & Marsman, M. (2024). Simplifying Bayesian analysis of graphical models for the social sciences with easybgm: A user-friendly R-package. *advances.in/psychology*, *e66366*. doi: 10.56296/aip00010
- Martarelli, C. S., Baillifard, A., & Audrin, C. (2022). A trait-based network perspective on the validation of the french short boredom proneness scale. *European Journal of Psychological Assessment*, *39*, 390–399. doi: 10.1027/1015-5759/a000718346
- Marsman, M., van den Bergh, D., & Haslbeck, J. M. B. (2025). Bayesian analysis of the ordinal Markov random field. *Psychometrika*, *90*, 146–182. doi: 10.1017/psy.2024.4
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